Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 5A nr. 5.5

$$
\begin{aligned}
&\int_0^{2\pi} \sin^2 \frac{x}{2} dx \\
&= \int_0^{2\pi} \frac{1-\cos x}{2} dx \\
&= \frac{1}{2} \int_0^{2\pi} (1-\cos x) dx \\
&= \frac{1}{2} \left(\int_0^{2\pi} 1 dx - \int_0^{2\pi} \cos x dx \right) \\
&= \frac{1}{2} \left([x]_0^{2\pi} - [\sin x]_0^{2\pi} \right) \\
&= \frac{1}{2} \left((2\pi - 0) - (\sin(2\pi) - \sin(0)) \right) \\
&= \frac{1}{2} (2\pi - 0) \\
&= \pi
\end{aligned}
$$

References